

LUDINGTON HARBOR, MI, USA

WMO: 997261

Lat: 43.947N

Lon: 86.441W

Elev: 589

StdP: 14.39

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.5	8.2	N/A	N/A	N/A	N/A	N/A	N/A	26.5	28.9	24.0	32.0	8.7	300	0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	80.6	N/A	78.5	N/A	76.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	9.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.6	20.2	17.9	DB	-1.8	85.4	4.9	4.9	-5.3	88.9	-8.2	91.8	-10.9	94.6	-14.5	98.2	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.0	24.7	24.7	32.7	44.3	54.3	64.0	68.1	67.7	61.3	50.4	39.7	30.6	(6)	
	DBStd	17.46	9.17	8.45	9.91	7.94	7.95	6.15	5.52	4.86	6.82	7.87	8.12	8.01	(7)	
	HDD50	3307	783	709	546	210	44	1	0	0	2	90	322	600	(8)	
	HDD65	6935	1248	1129	1003	623	348	92	33	27	152	457	758	1065	(9)	
	CDD50	2211	0	0	9	38	176	420	561	549	340	104	14	0	(10)	
	CDD65	365	0	0	0	1	15	62	129	111	41	6	0	0	(11)	
	CDH74	1217	0	0	0	2	69	237	464	311	117	17	0	0	(12)	
	CDH80	82	0	0	0	0	5	20	34	17	6	0	0	0	(13)	
(14) Wind	WSAvg	7.6	8.8	8.2	7.6	8.1	7.5	6.4	6.2	6.1	6.5	7.8	8.9	8.8	(14)	
(15) Precipitation	PrecAvg	33.4	2.0	1.5	2.0	2.9	2.6	3.0	2.1	4.0	4.1	3.4	3.1	2.7	(15)	
	PrecMax	42.3	4.5	3.6	7.1	4.6	5.5	8.3	4.3	9.4	10.7	7.7	7.3	6.4	(16)	
	PrecMin	24.3	0.4	0.2	0.3	1.0	0.4	0.2	0.6	0.3	0.2	1.0	0.6	0.9	(17)	
	PrecStd	4.6	1.1	0.9	1.5	1.0	1.4	1.8	1.1	2.2	2.4	1.5	1.5	1.4	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.5	48.1	68.7	72.7	80.8	83.0	84.1	82.1	80.8	76.7	63.7	55.9	(19)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)	
	2%	DB	43.6	41.9	60.2	65.9	75.7	79.1	80.4	79.3	77.2	71.6	59.3	47.2	(21)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)	
	5%	DB	40.4	38.2	52.2	60.7	71.0	76.8	78.4	77.2	74.8	66.6	55.1	44.2	(23)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)	
	10%	DB	37.7	35.7	46.4	56.3	66.8	74.3	76.7	75.6	72.4	62.5	51.8	41.5	(25)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)	
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)	
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)	
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	10.1	12.9	13.9	14.7	13.9	11.8	11.7	13.7	12.3	10.4	8.5	(35)	
		MCDBR	12.6	12.1	17.5	20.0	19.2	16.5	12.6	11.4	14.5	15.8	15.0	12.2	(36)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)	
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.293	0.312	0.335	0.367	0.397	0.409	0.411	0.402	0.384	0.349	0.317	0.289	(40)	
	taud		2.438	2.372	2.394	2.334	2.316	2.325	2.329	2.351	2.382	2.468	2.497	2.487	(41)	
	Ebn at Noon		266	279	286	285	278	274	272	270	266	263	255	257	(42)	
	Edh at Noon		23	30	33	38	40	40	39	37	33	27	22	20	(43)	
(44) All-Sky Solar Radiation	RadAvg		350	626	1120	1503	1809	1990	2023	1717	1344	812	453	279	(44)	
	RadStd		43	72	116	141	157	132	91	108	114	96	67	40	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page